

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

One-variable data: Distributions and measures of center and spread

01

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

Value	Frequency
1	a
2	$2a$
3	$3a$
4	$2a$
5	a

The frequency distribution above summarizes a set of data, where a is a positive integer. How much greater is the mean of the set of data than the median?

- A. 0
- B. 1
- C. 2
- D. 3

Data set A consists of 10 positive integers less than 60. The list shown gives 9 of the integers from data set A.

43, 45, 44, 43, 38, 39, 40, 46, 40

The mean of these 9 integers is 42. If the mean of data set A is an integer that is greater than 42, what is the value of the largest integer from data set A?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

- A. 20
- B. 24
- C. 32
- D. 36

Weight (pounds)	Frequency
13	12
14	8
15	5
16	7
17	9
18	10
19	13
20	7

The frequency table summarizes a data set of the weights, rounded to the nearest pound, of 71 tortoises. A weight of 39 pounds is added to the original data set, creating a new data set of the weights, rounded to the nearest pound, of 72 tortoises. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- A. The mean of the new data set is greater than the mean of the original data set, and the median of the new data set is greater than the median of the original data set.
- B. The mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.
- C. The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- D. The mean of the new data set is less than the mean of the original data set, and the medians of the two data sets are equal.

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

Data set A consists of the heights of 75 buildings and has a mean of 32 meters. Data set B consists of the heights of 50 buildings and has a mean of 62 meters. Data set C consists of the heights of the 125 buildings from data sets A and B. What is the mean, in meters, of data set C?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

Data set F consists of 55 integers between 170 and 290. Data set G consists of all the integers in data set F as well as the integer 10. Which of the following must be less for data set F than for data set G?

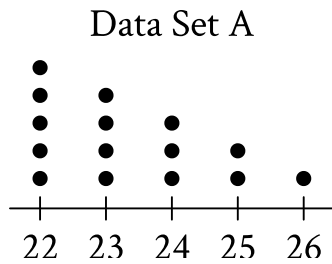
- I. The mean
 - II. The median
- A. I only
 - B. II only
 - C. I and II
 - D. Neither I nor II

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

- A. Median
- B. Mean
- C. Sum of the numbers
- D. Standard deviation



- The data for the dot plot are as follows:
 - 22: 5 dots
 - 23: 4 dots
 - 24: 3 dots
 - 25: 2 dots
 - 26: 1 dot

The dot plot represents the 15 values in data set A. Data set B is created by adding 56 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

- A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.
- B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

The mean amount of time that the 20 employees of a construction company have worked for the company is 6.7 years. After one of the employees leaves the company, the mean amount of time that the remaining employees have worked for the company is reduced to 6.25 years. How many years did the employee who left the company work for the company?

- A. 0.45
- B. 2.30
- C. 9.00
- D. 15.25

